

Ironhack, 2025

TripAdvisor Restaurants Analysis & Performance: **FRANCE & THE UK**



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Date:
13-02-2026

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INTRODUCTION, DATA
OVERVIEW &
OBJECTIVES

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PREDICTION WEB

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ANALYSIS, HYPOTHESIS &
INSIGHTS

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ENRICHMENT



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INTRODUCTION

Problem



CUISINE



VEGETARIAN



LANGUAGE

Objective

- Identify key drivers of restaurant ratings (e.g., cuisine, price range, vegetarian options).
- Compare the characteristics of restaurants and customer behavior.

Questions

- Are there significant differences in restaurant characteristics?
- How does vegetarian adoption influence restaurant ratings?

DATA OVERVIEW



Data Source

TripAdvisor European Restaurants, 2021
(Kaggle)

Preparation

Data Selection

- Focused on restaurants with:
 - Valid ratings and review counts
 - Clearly defined country and cuisine information
- Restricted analysis to **France** and the **UK** for cross-country comparison

RAW

1,083,397 rows

42 columns

CLEANED

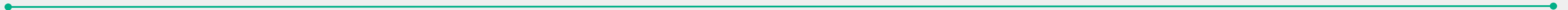
3,26,952 rows

29 columns

- Handled missing and inconsistent values
- Converted categorical variables to standard formats
- Removed duplicate values within list-based columns (cuisines, meals)



ANALYSIS HYPOTHESIS INSIGHTS



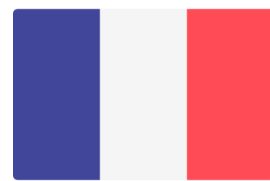


TOTAL RESTAURANTS



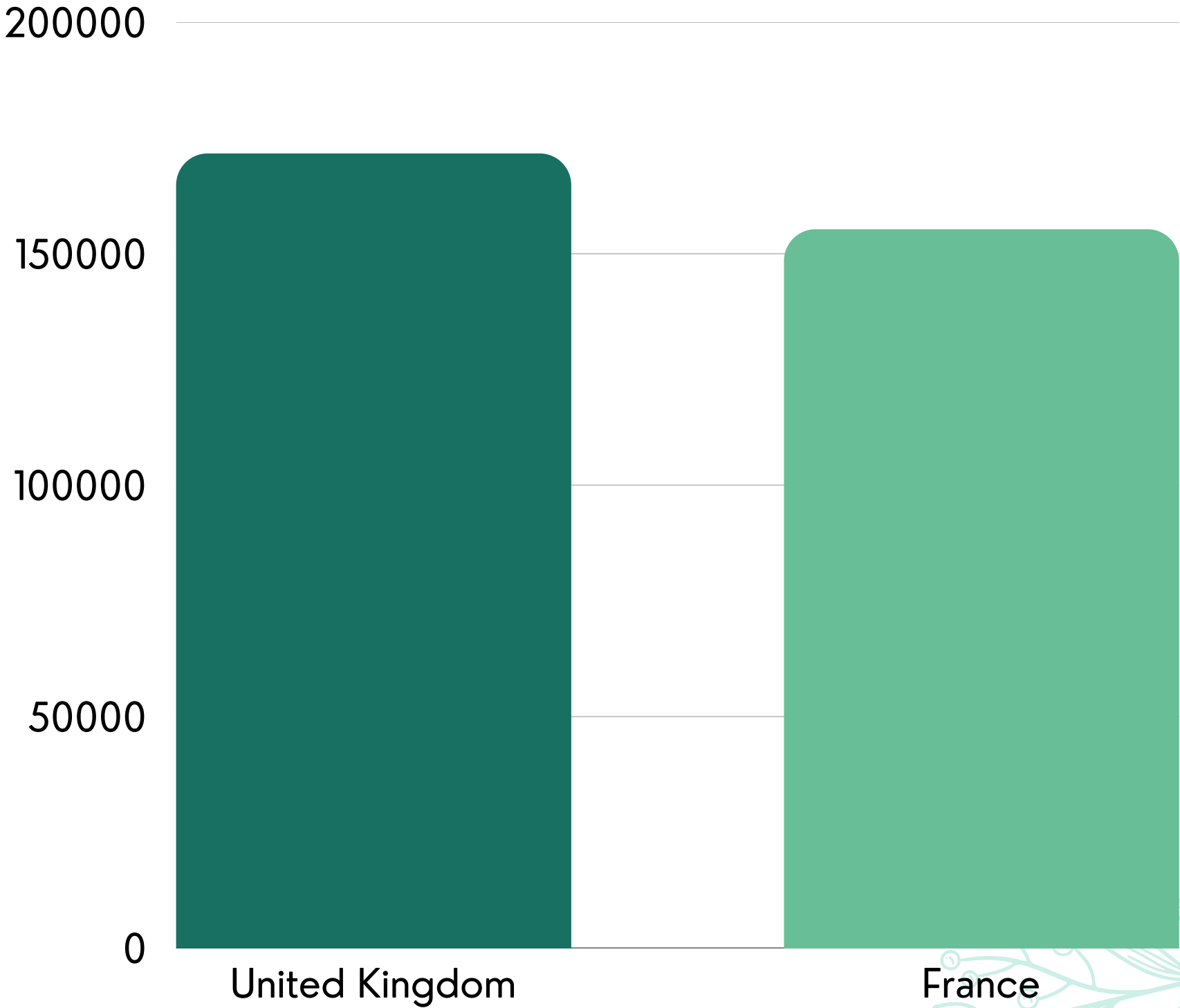
171664

~10% 



155288

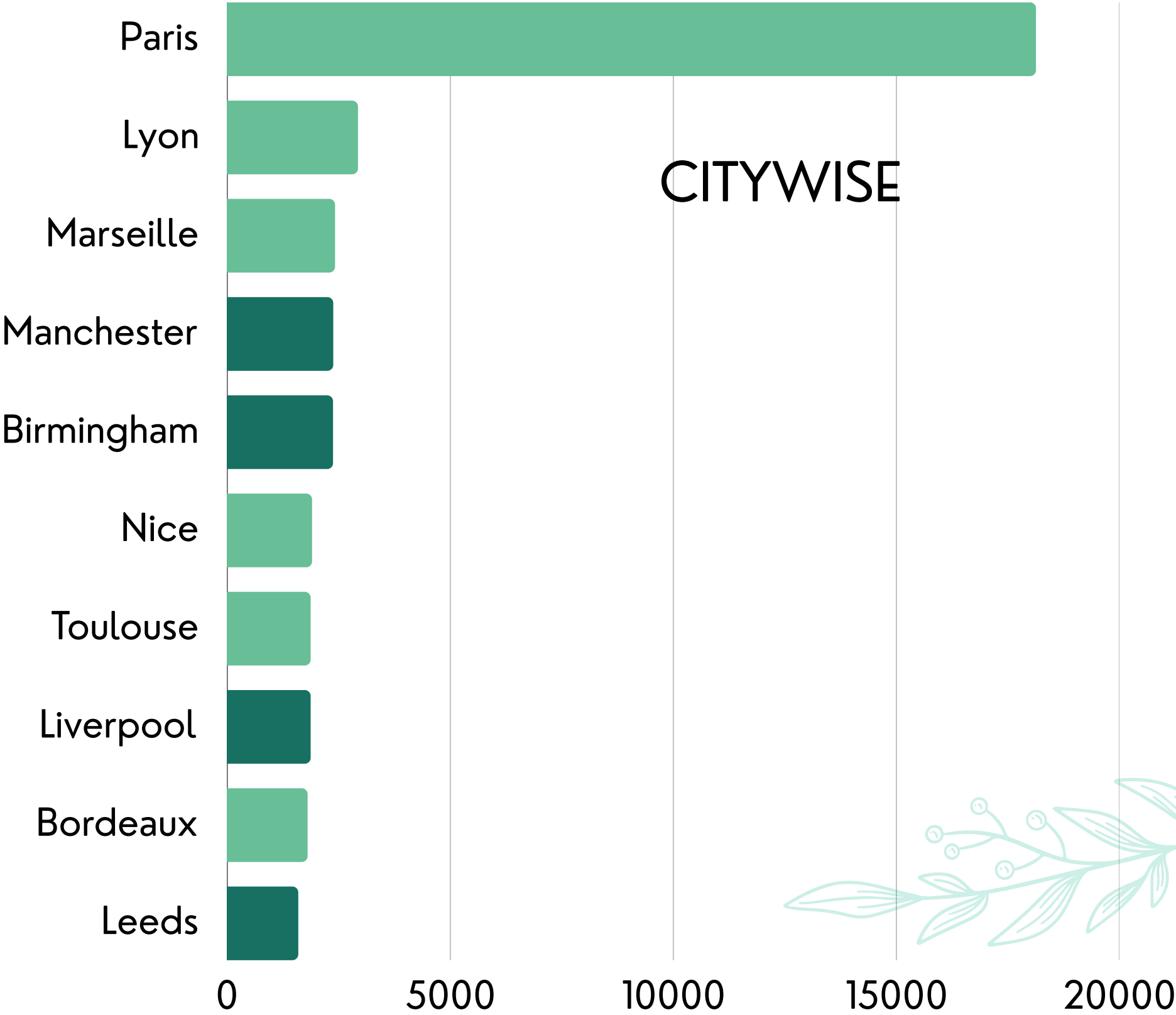
COUNTRYWISE

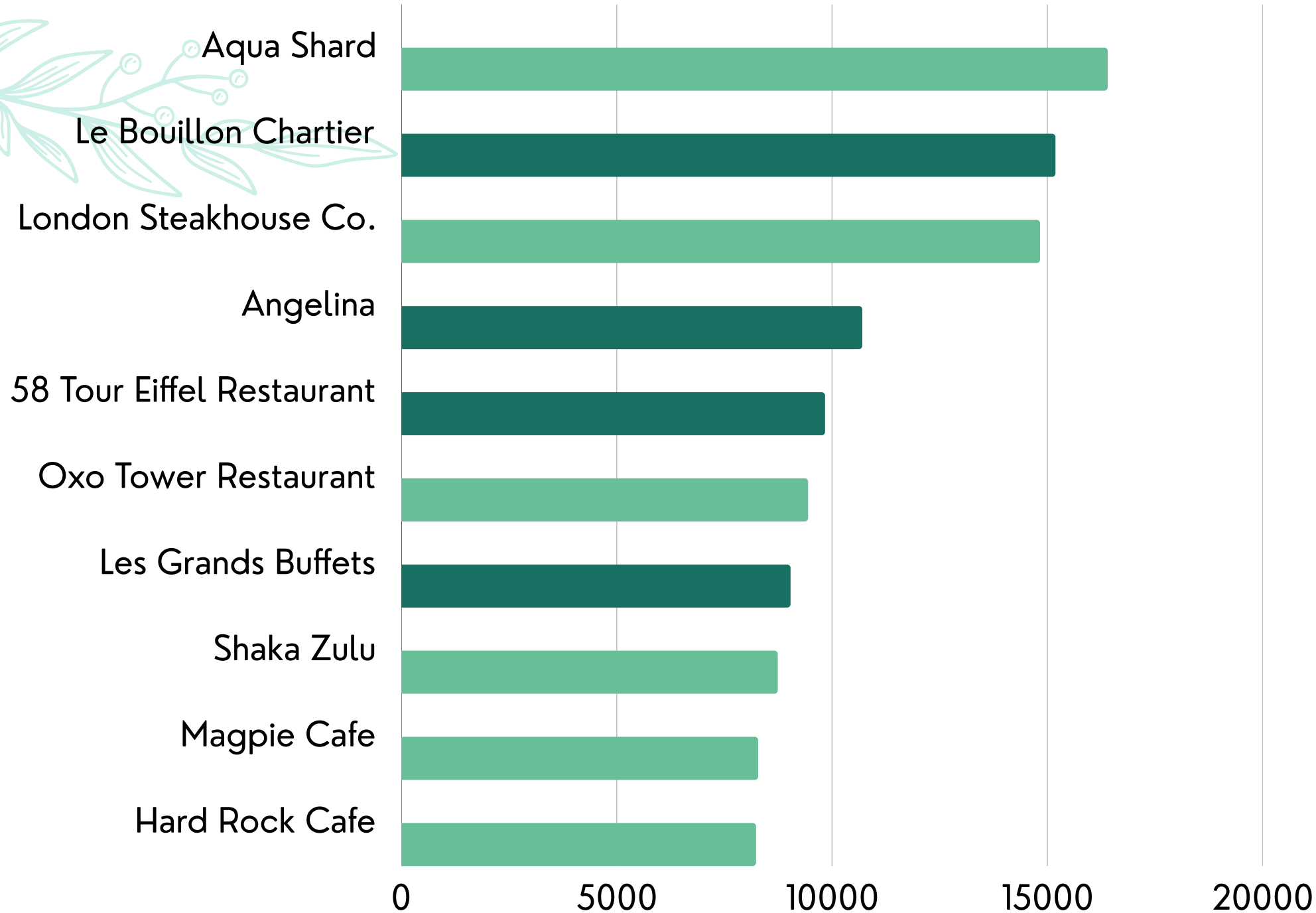


TOTAL RESTAURANTS

~12%
(of French Restaurants)

PARIS is in the lead





MOST REVIEWED RESTAURANTS

Aqua Shard (London, UK)

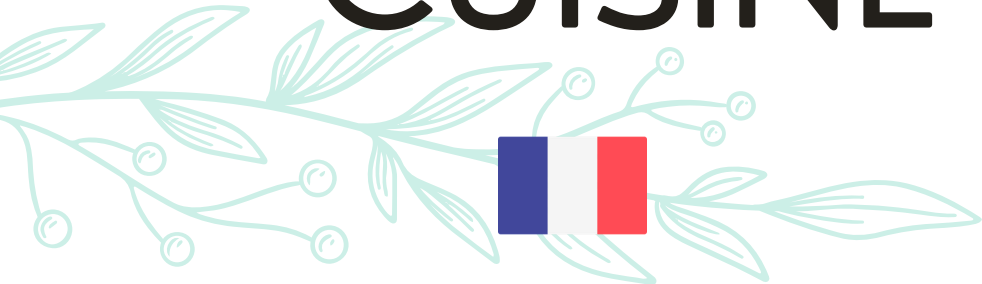


16,401 Review Count

- UK restaurants combine high ratings with large-scale popularity.
- French restaurants emphasize culinary excellence.



CUISINE LANDSCAPE



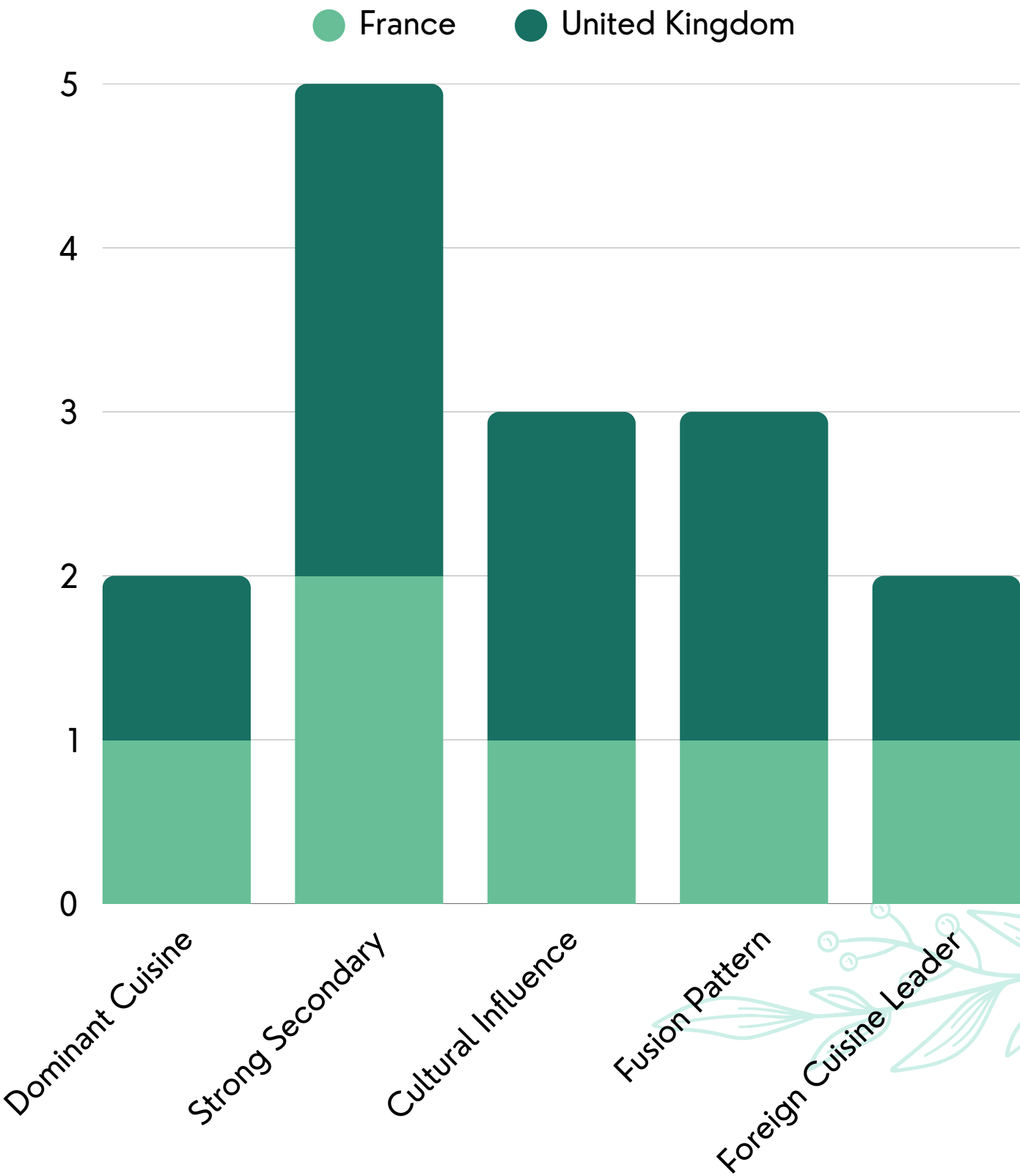
- **Dominant Cuisine:**
French
- **Strong Secondary:**
European, Italian
- **Cultural Influence:**
Traditional & regional
- **Fusion Pattern:**
European + French
- **Foreign Cuisine Leader:**
Italian



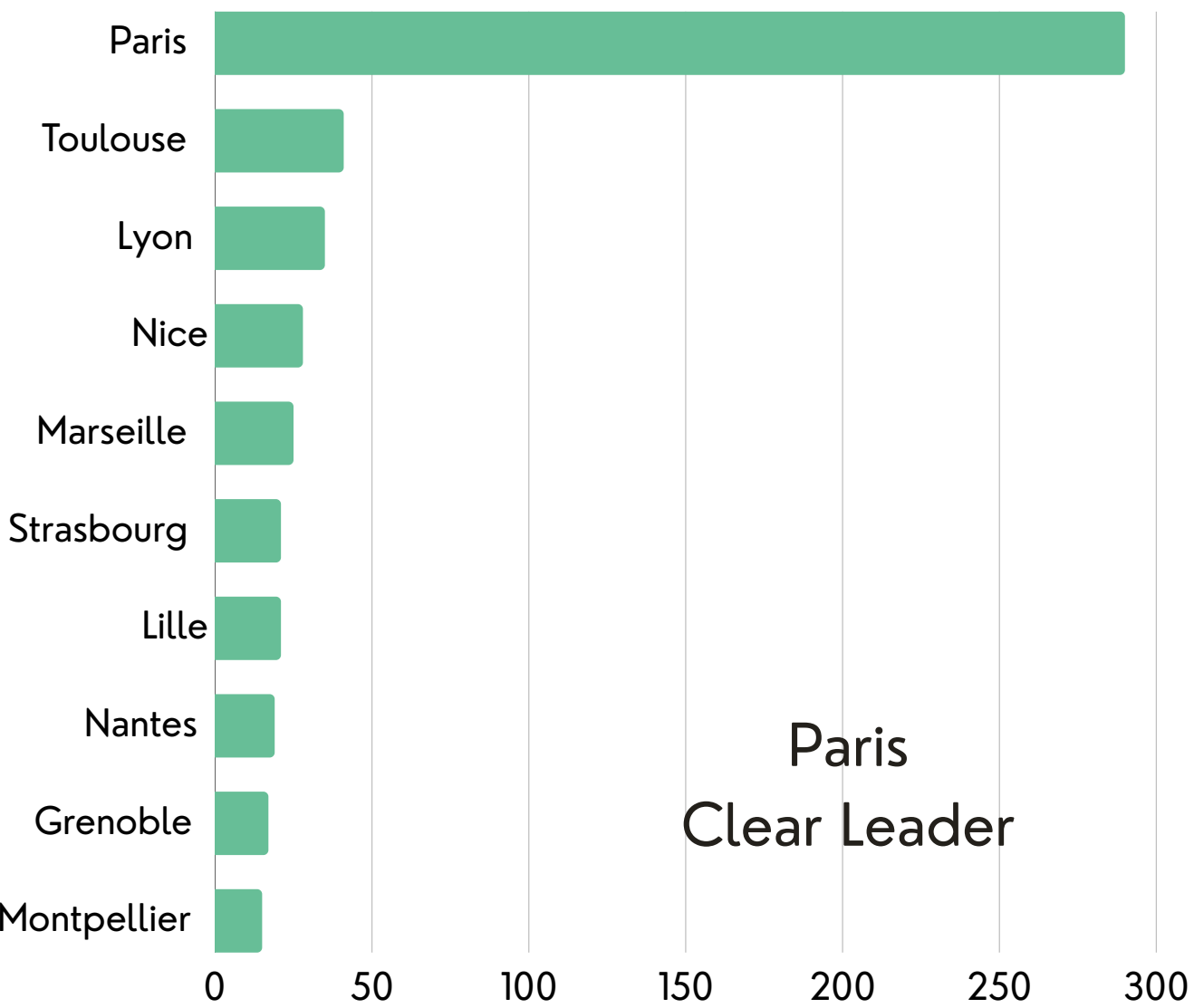
- **Dominant Cuisine:**
British
- **Strong Secondary:**
Cafe, Pub, Indian
- **Cultural Influence:**
Multicultural
- **Fusion Pattern:**
British + Pub / Bar
- **Foreign Cuisine Leader:**
Indian



INSIGHT: UK shows strong multicultural & foreign cuisine presence

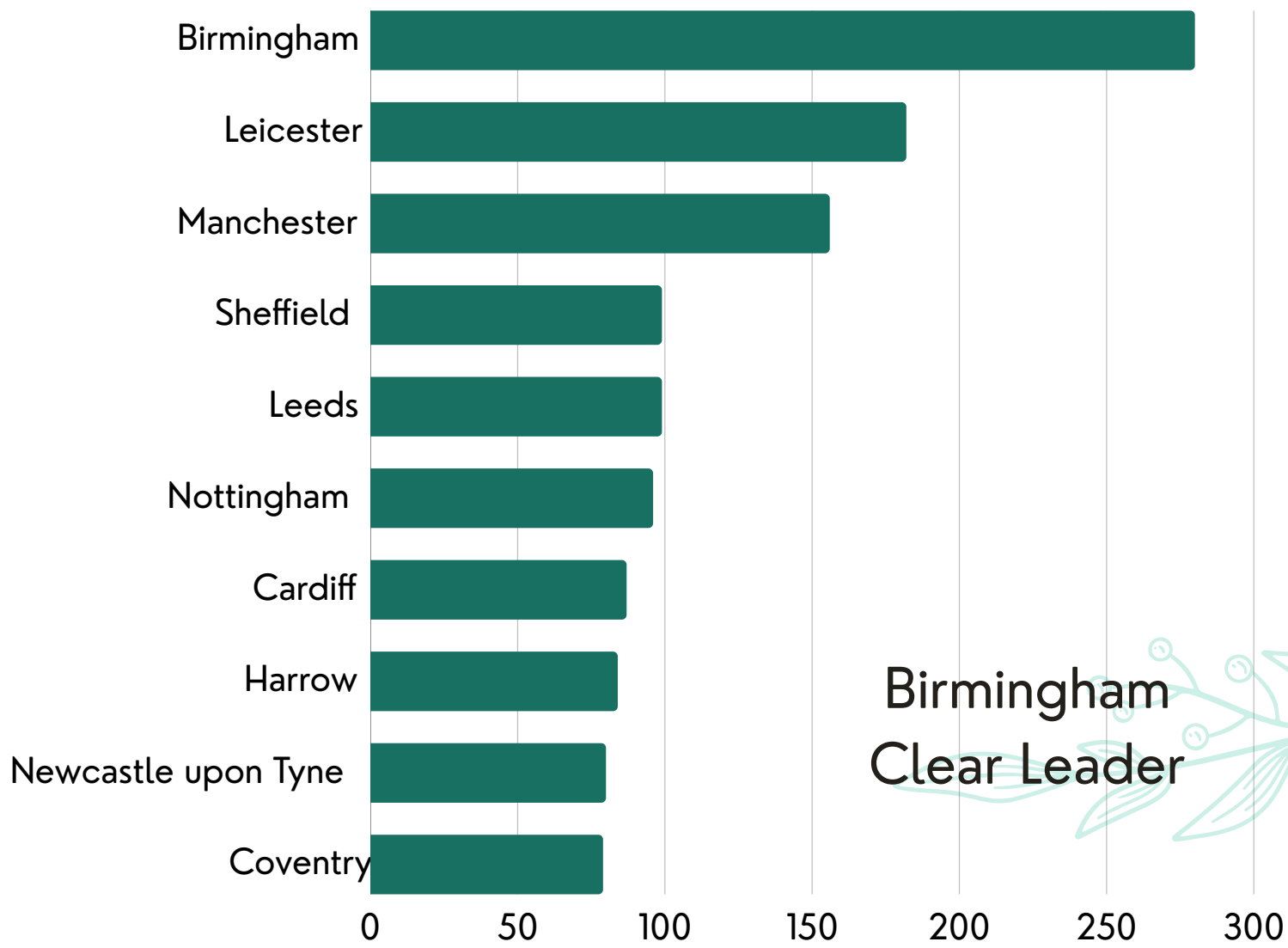


INDIAN CUISINE DISTRIBUTION




11.2%


88.8%

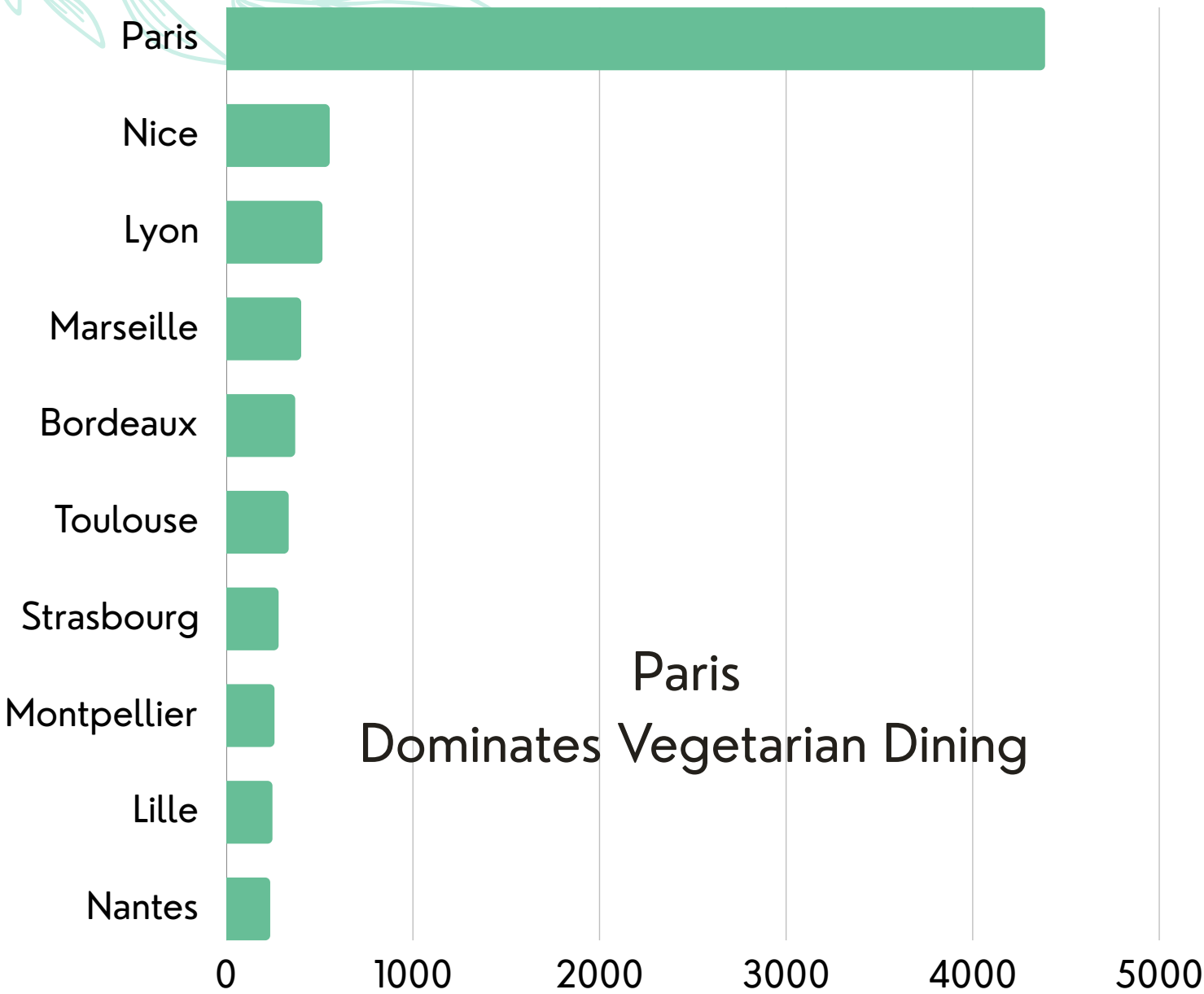


VEGETARIAN CUISINE DISTRIBUTION

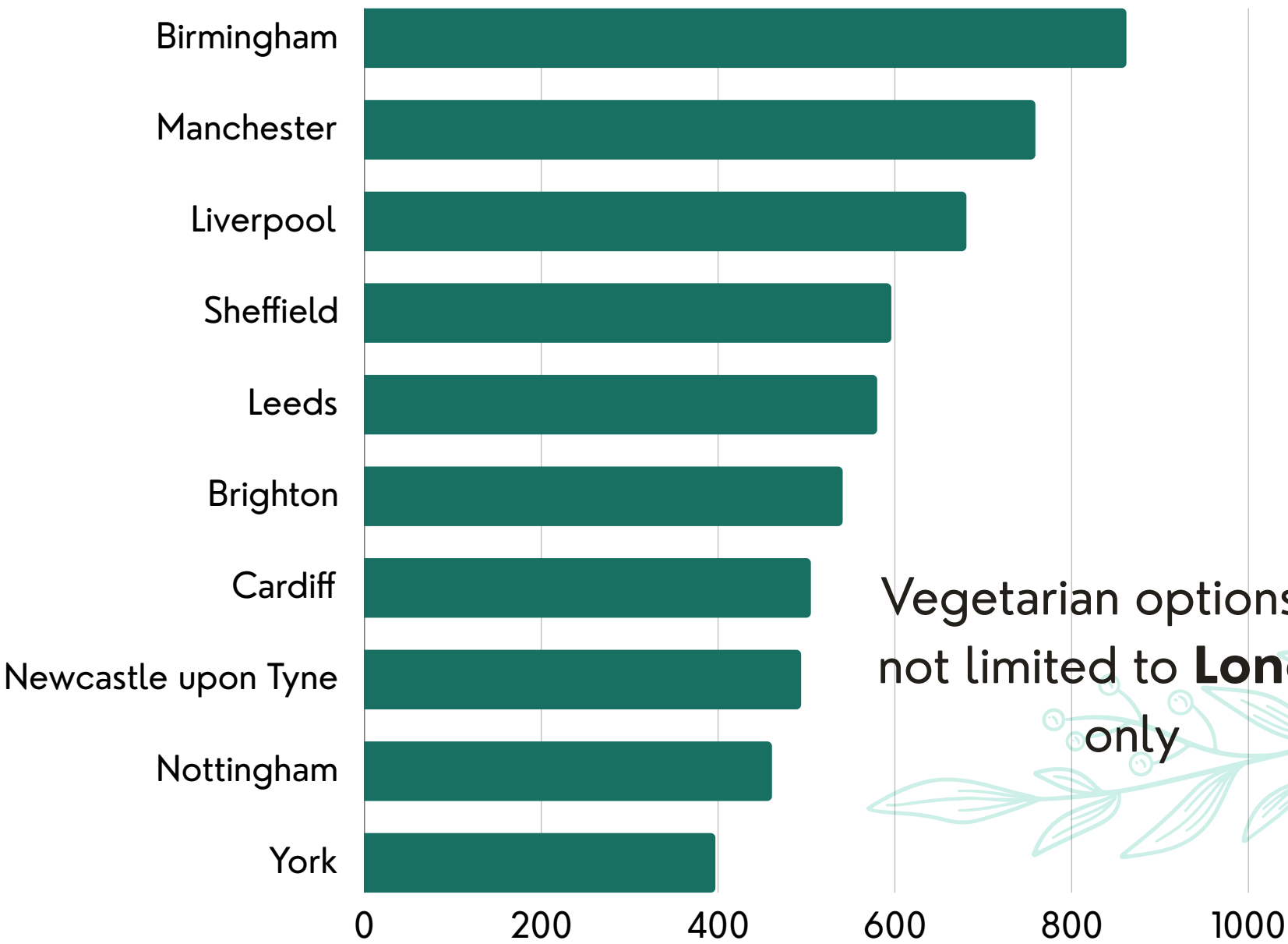


16.7%

44.2%



Paris
Dominates Vegetarian Dining



Vegetarian options are
not limited to **London**
only

ENGLISH LANGUAGE DISTRIBUTION

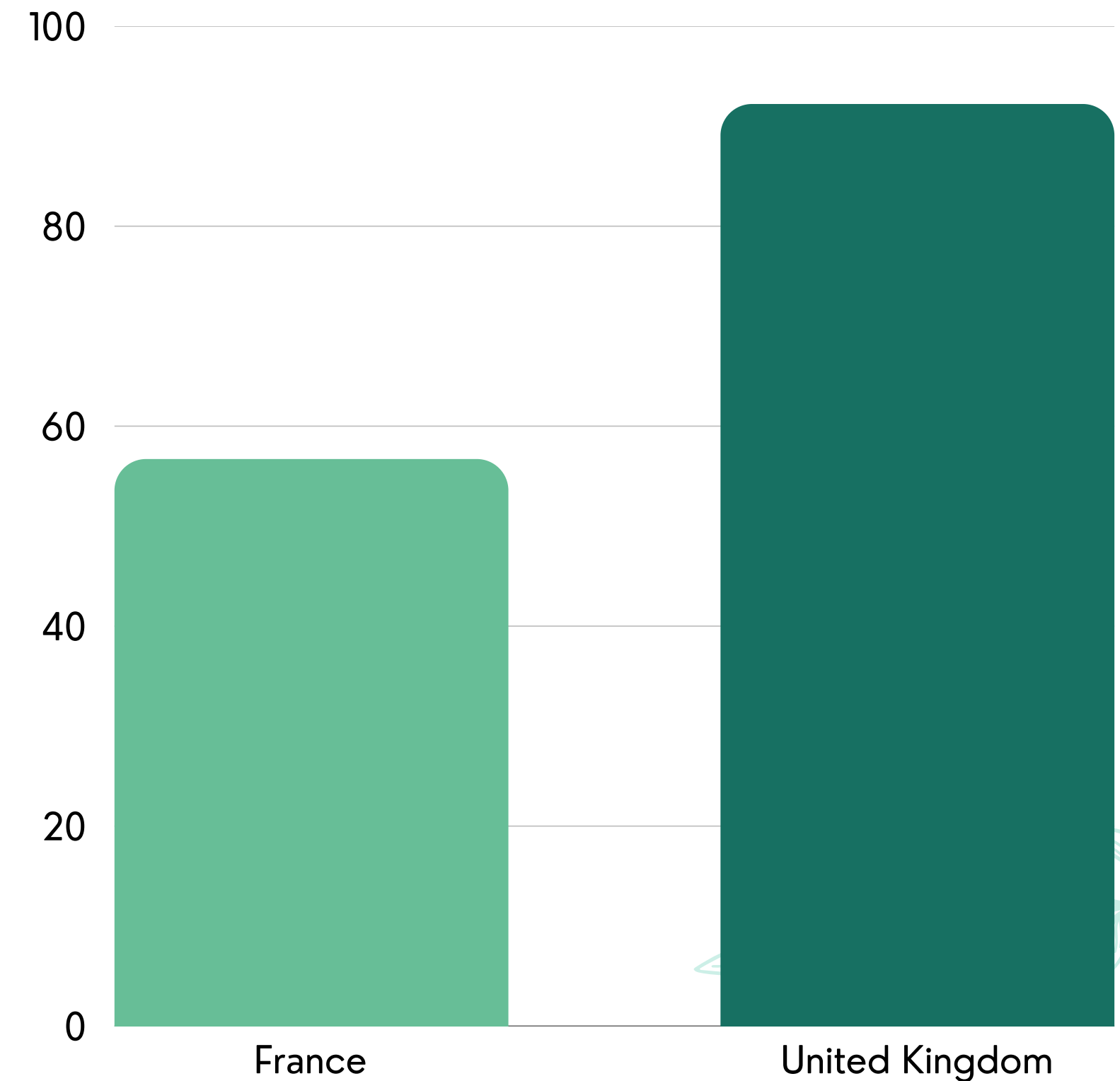


57%

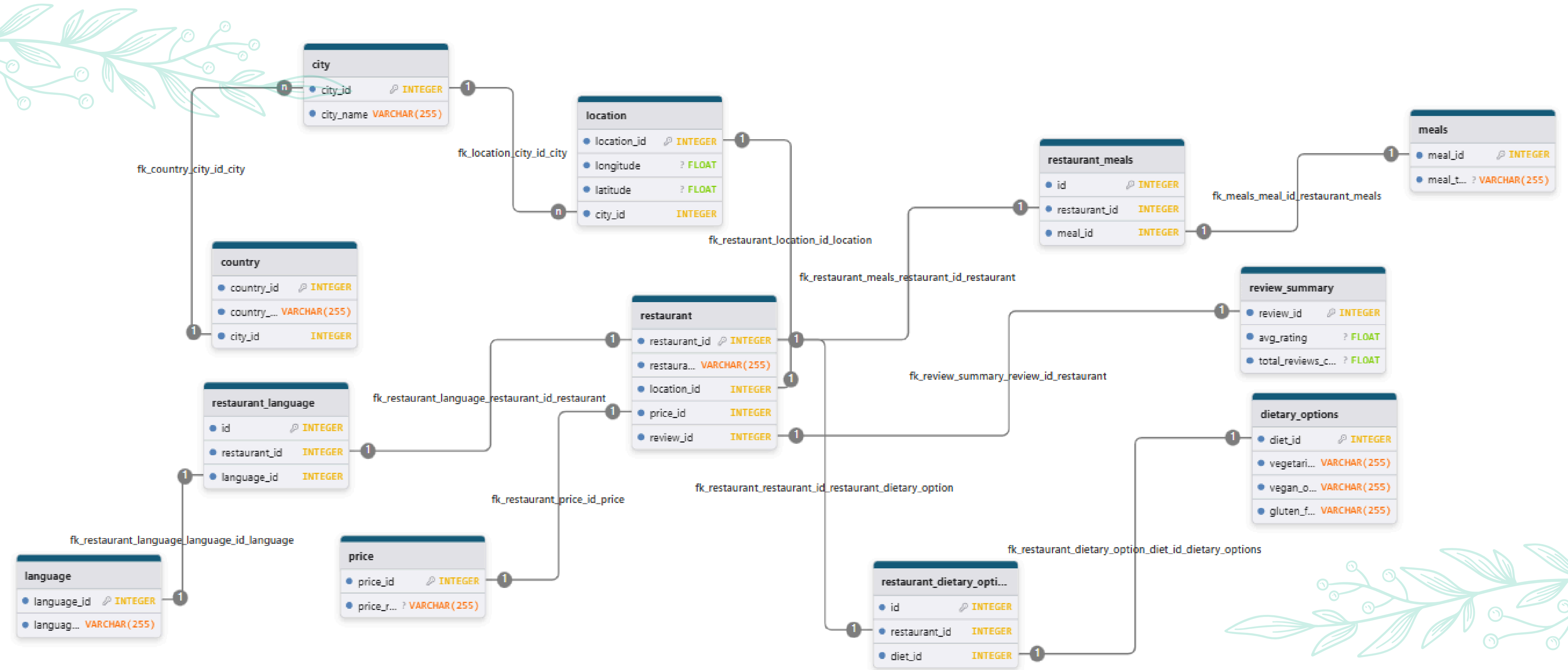


92%

- France: English is widely used but not dominant.
- UK: English is the default spoken language for restaurants.
- The UK is nearly universal, while France shows selective adoption driven by tourism.



ENTITY - RELATIONSHIP DIAGRAM (ERD)



Vegetarian-Friendly Restaurants

```
SELECT r.restaurant_name
FROM restaurant r
JOIN restaurant_dietary_option rdo
    ON r.restaurant_id = rdo.restaurant_id
JOIN dietary_options d
    ON rdo.diet_id = d.diet_id
WHERE d.vegetarian_friendly = 'Yes';
```

Restaurants with more than one language

```
SELECT r.restaurant_name,
    COUNT(r1.language_id) AS language_count
FROM restaurant r
JOIN restaurant_language r1
    ON r.restaurant_id = r1.restaurant_id
GROUP BY r.restaurant_name
HAVING language_count > 1;
```

DATABASE SCHEMA

Restaurant by City

```
SELECT c.city_name, COUNT(r.restaurant_id) AS total
FROM restaurant r
JOIN location l ON r.location_id = l.location_id
JOIN city c ON l.city_id = c.city_id
GROUP BY c.city_name
ORDER BY total DESC;
```

Restaurant by Country

```
SELECT co.country_name, COUNT(r.restaurant_id) AS total
FROM restaurant r
JOIN location l ON r.location_id = l.location_id
JOIN city c ON l.city_id = c.city_id
JOIN country co ON c.city_id = co.city_id
GROUP BY co.country_name
ORDER BY total DESC;
```





HYPOTHESIS EXPLANATION





VEGETARIAN ADOPTION

Is there a statistically significant difference in the adoption of vegetarian-friendly restaurants between France and the UK?

Null hypothesis (H_0):

Vegetarian adoption is independent of country (France and the UK have similar proportions)

Alternative hypothesis (H_1):

Vegetarian adoption differs by country

Statistical Test:

Chi-square test of independence

Result: $p < 0.05$



ENGLISH LANGUAGE DISTRIBUTION

Is there a statistically significant difference in the proportion of restaurants using English as the default language between France and the UK?

Null hypothesis (H_0):

There is no association between country (France vs. the UK) and English-language usage in restaurants

Alternative hypothesis (H_1):

There is an association between country and English-language usage.

Statistical Test:

Chi-square test of independence

Result: $p < 0.05$



RESTAURANT RATING PREDICTION



PREDICTION GOAL:

Can we forecast a restaurant's category (High, Medium, Low) based on its characteristics?

LOW
(< 4.0)



MEDIUM
($4.0 - 4.4$)



HIGH
(≥ 4.5)



DATA PREPROCESSING

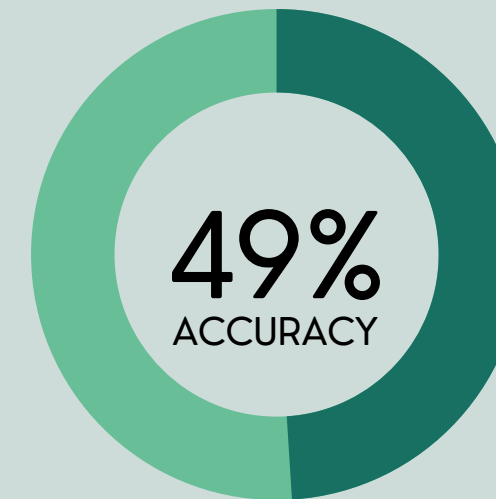
One-Hot Encoding
for Categorical
Standard Scaling
for Numerical



MODEL IMPLEMENTATION

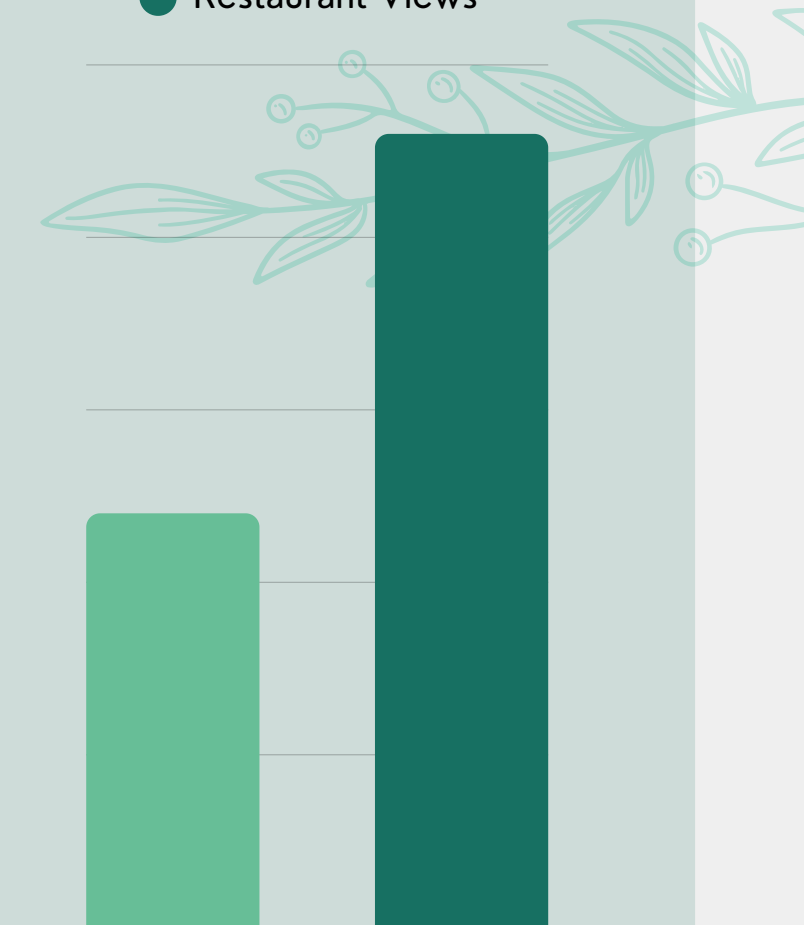
- ✓ Logistic Regression
- 🌳 Decision Tree
- 🌲 Random Forest ★
- ⚡ Gradient Boosting

MODEL PERFORMANCE & INSIGHTS



RECALL METRIC:

59%



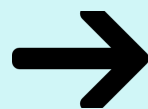
STRONG PREDICTORS

GOOGLE PLACES API ENRICHMENT

DATA ENRICHMENT GOAL & PROCESS

To enrich the dataset using the Google Places API to obtain up-to-date ratings, geolocation, and popularity metrics for Indian restaurants in France and the UK.

Search Query
Indian restaurant



TEXT
SEARCH
API

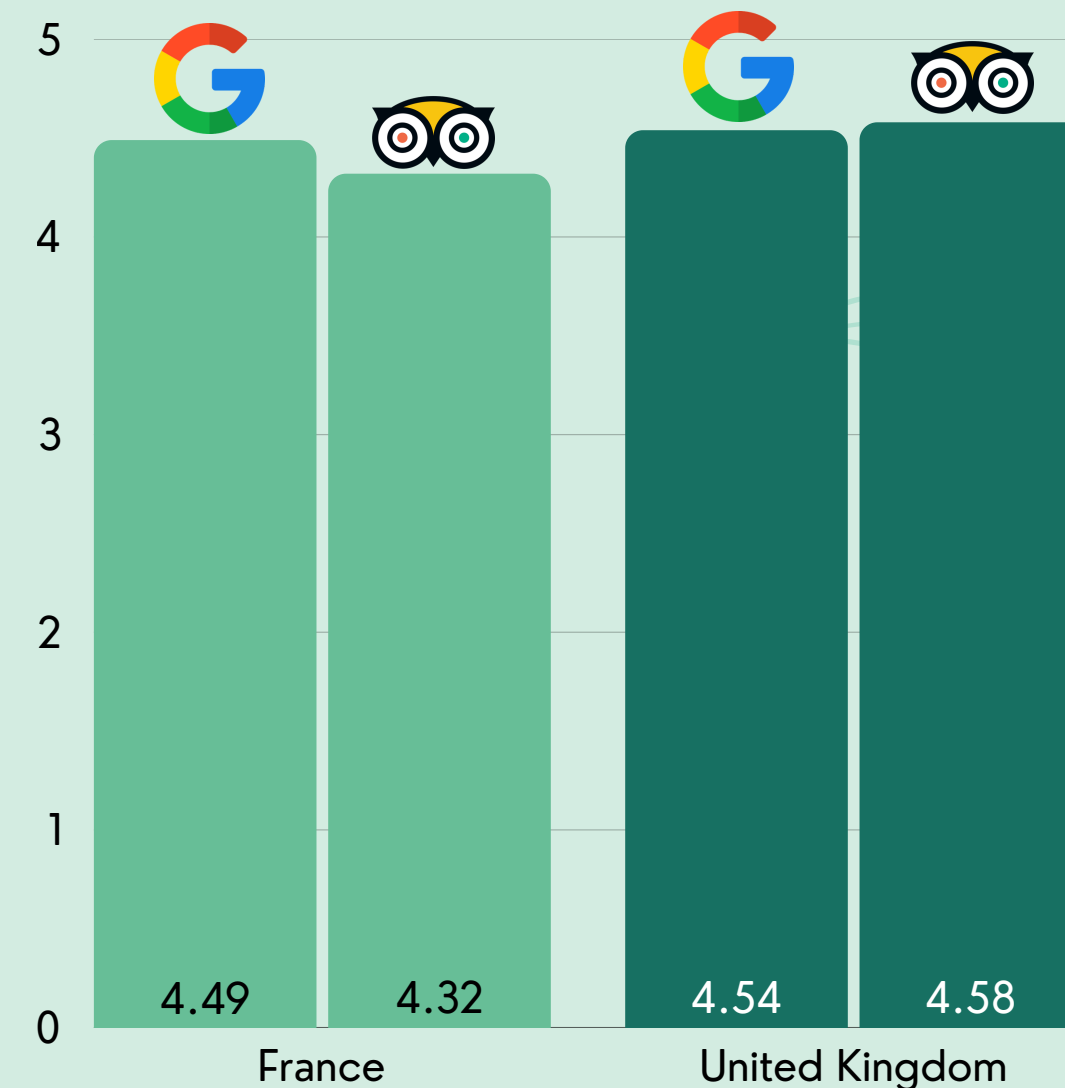


Paris, Lyon,
Marseille, Toulouse



London, Manchester,
Birmingham, Leicester

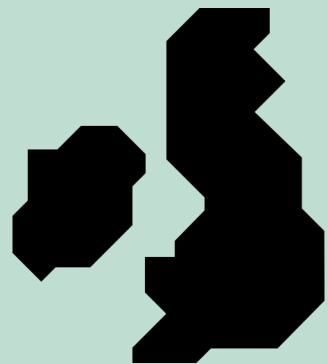
RATING COMPARISON: GOOGLE vs TRIPADVISOR



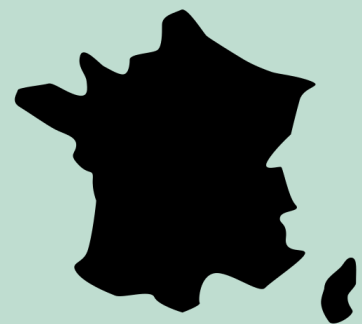
MAIN RESULTS

CULTURAL LANDSCAPE

UK



- Higher cuisine diversity & multicultural influence
- Higher proportion of vegetarian-friendly restaurants
- More prevalent Indian cuisine (stronger cultural integration and demand)



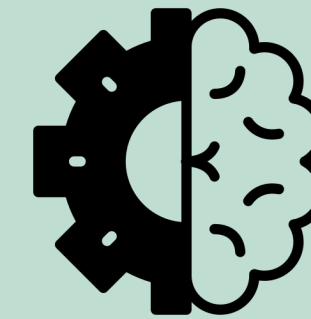
FRANCE

- Stronger traditional European cuisine dominance



RATING DRIVERS

- Popularity strongly influences ratings
- Total review count is a key predictor of high ratings.



TECHNICAL INSIGHTS

- Machine learning achieved
 - ~50% accuracy
 - Distinguishes high & low-rated restaurants.
 - Medium-rated restaurants: Rating overlap
- API enrichment enhances insights
 - External data improved spatial and popularity analysis.
 - Importance of balancing data quality and quantity.

CHALLENGES FACED



Data quality & missing values

Extensive cleaning needed due to missing, inconsistent & duplicate entries.



Cuisine classification

Restaurants listing multiple cuisines made categorization complex.



Subjectivity of ratings

Restaurant ratings reflect personal preferences, making objective comparison.



Cross-country difference

Differences in cultural dining norms between France and the UK added complexity to interpretation.

KEY LEARNINGS



Data preparation is critical

Cleaning & structuring data significantly impacted analysis quality.



Multicultural markets are more inclusive

The UK offers a broader range of dining options for international & vegetarian travelers.



Ratings are multifactorial

Restaurant ratings are influenced by cuisine type & cultural context.



Statistical testing adds confidence

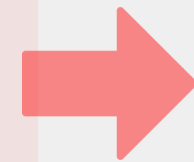
Hypothesis testing validated insights beyond visual analysis

MAJOR OBSTACLE

Data Standardization Across Countries

? The Challenge

- Inconsistent data in France & the UK.
- Non-standardized Cuisines, vegetarian options, and language indicators.



! Why it mattered

- Biased analyses & misleading results.
- Cross-country comparisons unreliable.



✓ How to solve it

- Standardized categories across countries.
- Clean & engineered data prior to analysis.

NEXT STEPS

EXPAND DATA COVERAGE



- Broaden geography & cities analyzed
- Incorporate temporal data for trend analysis

IMPROVE MODEL PERFORMANCE



- Handle imbalanced data (SMOTE)
- Test multi-class vs binary classification

SPATIAL & GEOGRAPHIC ANALYSIS



- Use latitude/longitude for clustering
- Identify restaurant density hotspots
- Analyze location impact on ratings

DEPLOY AS AN INTERACTIVE TOOL



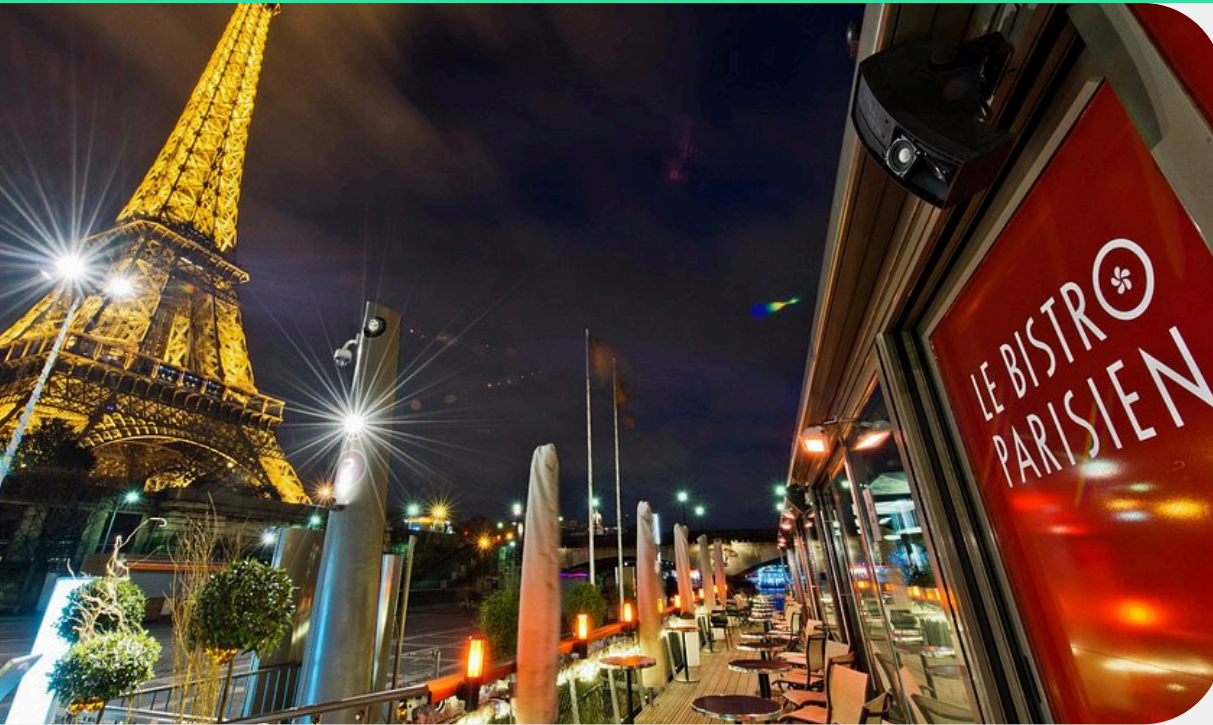
- Expand Streamlit app features
- Enable real-time predictions
- Integrate API

BROADEN DATA

ENHANCE ACCURACY

DERIVE INSIGHTS

BUILD APP



CONCLUSION



Project Goal

This project analyzed restaurant data from France and the United Kingdom to understand the drivers of customer ratings.



Key Findings

- Exploratory analysis & hypothesis testing revealed significant differences in dining culture, cuisine diversity, and vegetarian adoption.
- Popularity, cuisine type, & price range emerged as the most influential factors.



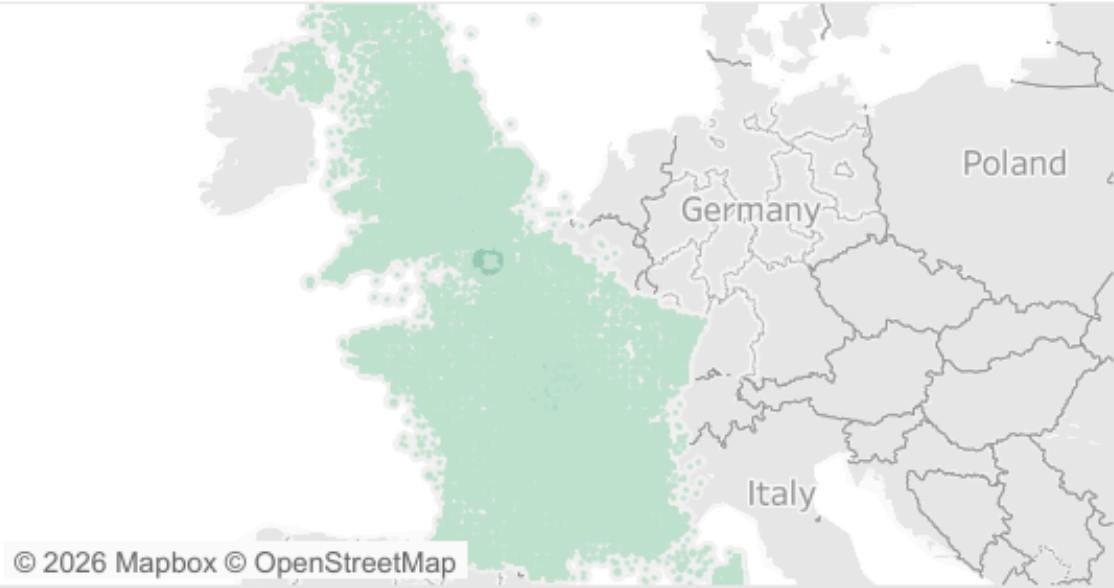
Methodology & Ethics

Ethical data practices and proper preprocessing were critical to ensuring reliable and interpretable results.

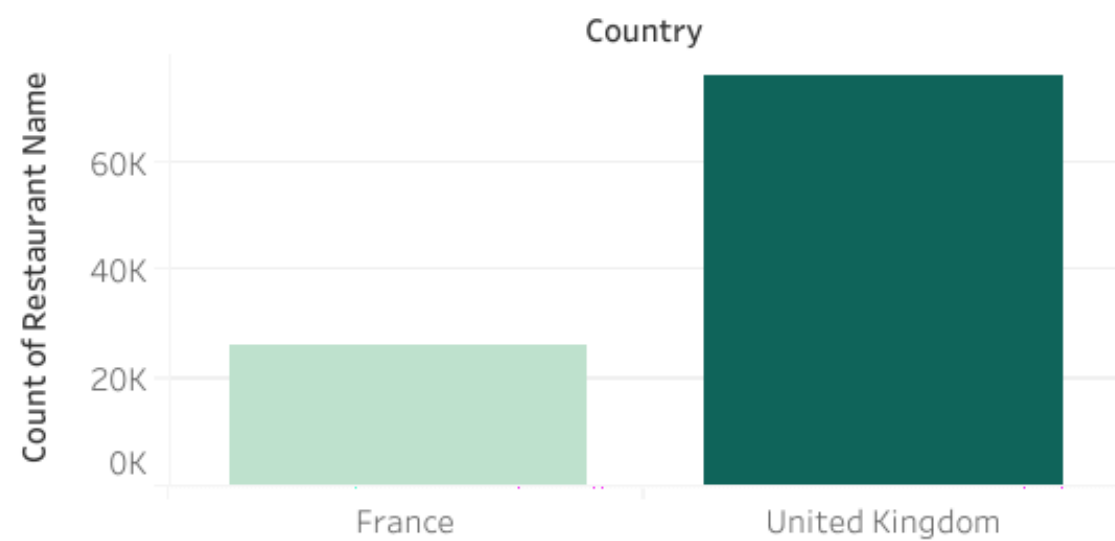
TABLEAU DASHBOARD

[CLICK HERE](#)

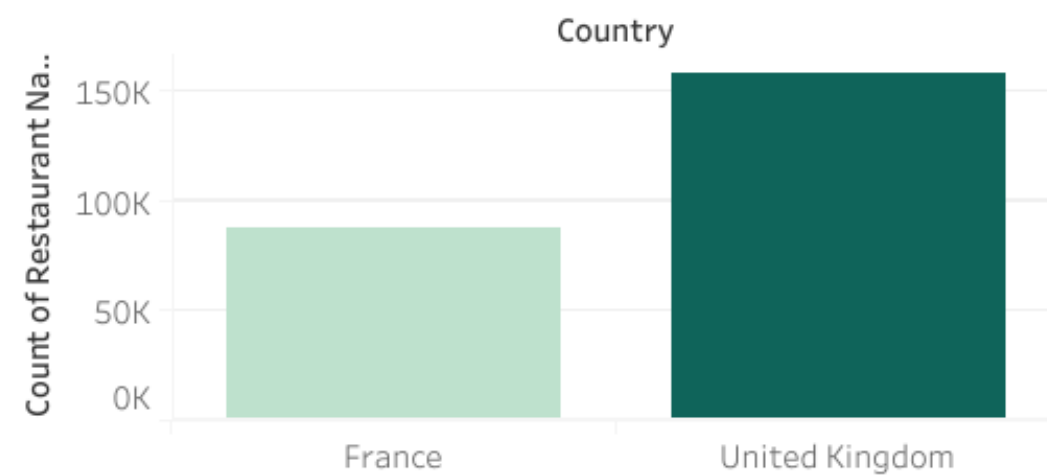
Restaurant Map



Vegetarian Restaurants by Country



English-Language Restaurants by Country

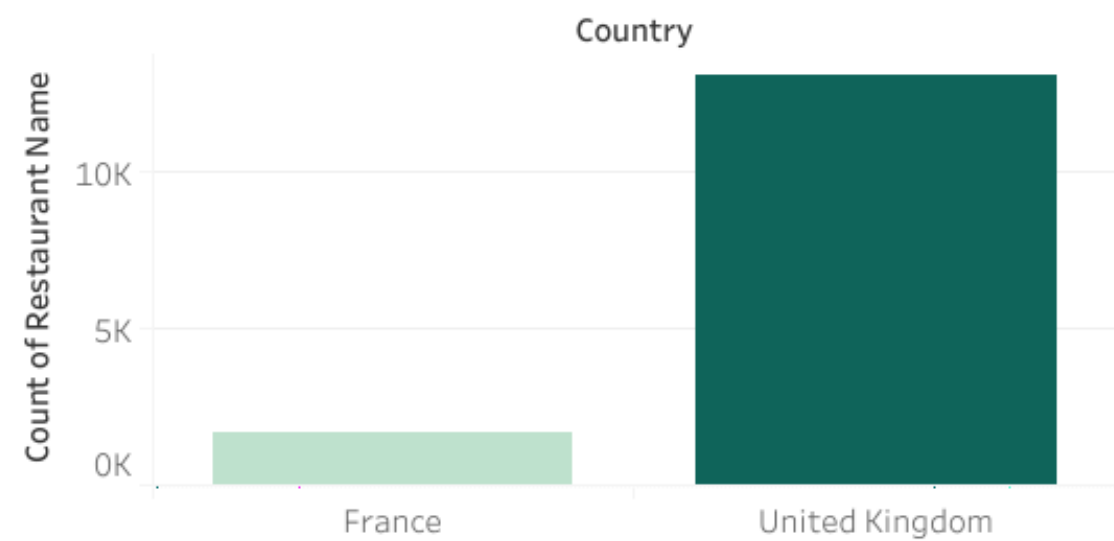


Avg Rating
1  10,479

Total Reviews Count
• 0
• 20,000
• 40,000
• 67,344

Restaurant Name
Limit

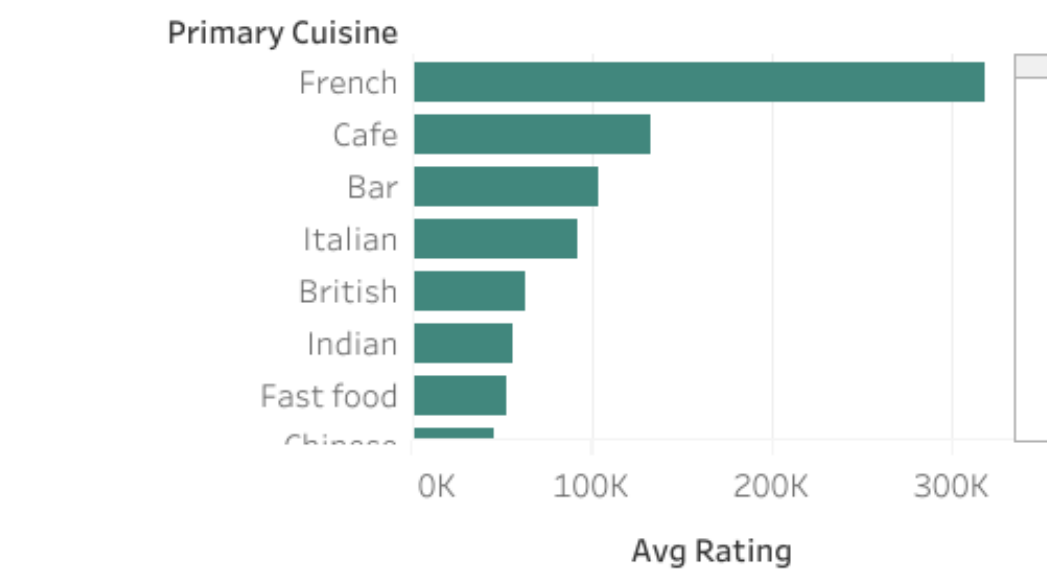
Indian Cuisine Distribution



Most Reviewed Restaurants by Country

Restaurant Name	Country	
	France	United Kingdom
Leon de Bruxelles	67,344	
Pizza Express	72	66,448
Buffalo Grill	57,585	458
McDonald's	41,644	10,547
Bella Italia	869	48,077
Total Reviews Count		

Average Rating by Cuisine



- Primary Cuisine**
- ☒ Afghani
 - ☒ African
 - ☒ Albanian
 - ☒ Algerian
 - ☒ American
 - ☒ Arabic
 - ☒ Argentinian
 - ☒ Armenian
 - ☒ Asian
 - ☒ Australian
 - ☒ Austrian
 - ☒ Azerbaijani
 - ☒ Balti
 - ☒ Bangladeshi
 - ☒ Bar
 - ☒ Barbecue



A QUICK DEMO

[CLICK HERE](#)



 Thank you for your
time & attention!



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